

MERIMBULA, Australia

WMO: 959290

Lat: 36.9078S

Lon: 149.8989E

Elev: 2

StdP: 101.30

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.2	-1.2	3.4	12.0	-0.1	3.7	10.3	9.3	15.6	8.3	14.9	1.4	220	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	27.1	20.6	25.7	20.2	24.7	19.7	22.0	25.4	21.2	24.3	20.7	23.7	5.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.4	24.0	20.1	14.8	23.4	19.4	14.2	22.8	64.4	25.5	61.8	24.3	59.6	23.7	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.8	6.8	DB	0.3	33.2	0.7	2.8	-0.2	35.2	-0.6	36.9	-1.0	38.5	-1.5	40.5	(4)
(5)			WB	-0.1	23.4	0.6	1.0	-0.6	24.1	-1.0	24.7	-1.3	25.3	-1.8	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	19.6	19.6	18.5	16.2	13.3	11.2	10.3	10.7	12.5	14.4	16.2	18.1	(6)	
	DBStd	3.91	2.17	2.05	2.08	1.92	1.79	1.68	1.61	1.85	2.23	2.47	2.32	2.38	(7)	
	HDD10.0	37	0	0	0	0	0	7	16	12	2	1	0	0	(8)	
	HDD18.3	1377	11	9	23	68	156	216	250	236	177	127	70	34	(9)	
	CDD10.0	1866	299	269	263	186	102	41	23	34	77	135	187	250	(10)	
	CDD18.3	166	52	45	28	4	0	0	0	0	1	3	7	26	(11)	
	CDH23.3	639	197	144	91	21	3	0	0	1	16	36	31	99	(12)	
	CDH26.7	86	24	9	8	4	0	0	0	0	3	12	6	20	(13)	
(14) Wind	WSAvg	2.7	3.1	2.9	2.6	2.4	2.4	2.4	2.4	2.7	2.8	3.0	3.2	3.1	(14)	
(15) Precipitation	PrecAvg	833	77	79	91	83	65	69	41	41	56	69	86	67	(15)	
	PrecMax	1444	181	392	256	447	280	315	133	158	144	327	270	282	(16)	
	PrecMin	316	14	2	7	5	1	2	1	4	5	10	5	2	(17)	
	PrecStd	267	53	81	71	109	65	79	39	36	41	75	69	59	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	28.0	27.1	25.7	23.1	19.7	20.1	21.9	25.3	27.1	26.4	28.9	(19)	
		MCWB	21.6	22.4	21.4	17.5	15.6	12.7	11.8	12.5	14.6	16.4	19.1	20.6	(20)	
	2%	DB	26.7	26.1	25.3	23.2	21.0	18.0	17.9	19.1	21.6	23.1	23.7	25.5	(21)	
		MCWB	21.2	21.0	20.3	17.6	14.9	12.5	11.4	11.9	13.7	16.4	18.4	19.9	(22)	
	5%	DB	25.2	25.0	24.2	22.0	19.5	16.9	16.5	17.2	19.3	21.1	22.3	24.0	(23)	
		MCWB	20.5	20.4	19.6	17.2	14.2	12.0	11.0	11.5	13.6	16.0	17.7	19.4	(24)	
	10%	DB	24.2	24.1	23.1	20.9	18.2	15.8	15.4	16.1	17.9	19.6	21.2	22.8	(25)	
		MCWB	19.8	19.8	18.9	16.5	13.3	11.6	10.4	11.0	12.8	15.1	16.9	18.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	22.9	22.2	19.7	17.3	15.3	14.2	14.6	16.3	18.8	20.5	22.0	(27)	
		MCDB	27.5	26.4	25.6	23.7	20.1	17.2	17.0	18.5	20.6	22.7	24.0	26.3	(28)	
	2%	WB	21.9	22.0	21.2	18.7	16.1	13.8	12.8	13.4	15.3	17.7	19.4	20.8	(29)	
		MCDB	25.5	25.0	24.0	21.9	19.4	16.4	16.2	17.3	19.3	21.2	22.6	24.4	(30)	
	5%	WB	21.2	21.2	20.5	18.0	15.2	12.9	11.7	12.4	14.4	16.8	18.5	20.0	(31)	
		MCDB	24.4	24.1	23.3	20.9	18.5	15.5	15.3	16.0	18.3	20.2	21.5	23.2	(32)	
	10%	WB	20.5	20.5	19.6	17.1	14.1	12.1	10.9	11.5	13.5	15.9	17.6	19.2	(33)	
		MCDB	23.6	23.4	22.5	20.0	17.1	14.8	14.5	15.1	17.2	18.9	20.5	22.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	7.9	8.7	9.5	10.5	10.0	11.3	11.3	11.1	10.1	8.8	8.4	(35)	
		MCDBR	9.8	9.2	10.2	11.2	12.7	11.7	13.2	13.2	13.7	12.9	10.6	10.2	(36)	
	5% WB	MCWBR	5.8	5.3	6.0	6.5	7.5	7.0	7.7	7.6	7.5	7.1	6.2	5.9	(37)	
		MCWBR	8.9	8.1	9.0	9.7	11.2	9.5	11.7	11.9	12.2	11.0	9.3	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.422	0.384	0.351	0.331	0.315	0.303	0.293	0.305	0.334	0.347	0.369	0.376	(40)	
		taud	2.340	2.491	2.567	2.596	2.590	2.616	2.639	2.603	2.521	2.515	2.487	2.482	(41)	
	Ebn at Noon		913	926	921	880	837	821	856	897	923	953	959	962	(42)	
		Edh at Noon	133	110	94	81	70	64	66	78	97	106	114	116	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.53	5.62	4.75	3.60	2.68	2.10	2.46	3.30	4.54	5.60	6.18	6.69	(44)	
	RadStd		0.49	0.43	0.34	0.27	0.13	0.14	0.19	0.17	0.31	0.32	0.44	0.44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+45

Nomenclature: See separate page